QP CODE: 24900028

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS)
REGULAR EXAMINATION, NOVEMBER 2024
First Semester

Discipline Specific Core course- MG1DSCBTS100- NEW BIOLOGY FOR THE NEW CENTURY
(2024 Admission Onwards)

Time:1.5 Hours **Max. Marks: 50**

Part A

Multiple Choice Questions. Answer **All** questions.
Each question carries **1** mark

1. Cell (c) [K] [1]
2. Plasma Membrane (c) [U] [1]
3. Sample Preparation (d) [S] [2]
4. Monoclonal antibodies (c) [I] [2]
5. Brown eyes (c) [K] [3]
6. Amino acid (d) [K] [3]
7. Carry genetic information from DNA to ribosomes (a) [U] [4]
8. Golgi apparatus (b) [U] [5]
9. The variety of life in a particular habitat or ecosystem (b) [U] [6]
10. DNA Sequence (c) [K] [7]

(10 X 1 =10 Marks)

Part B

Short Answer Type Questions. Answer any **Six** questions.
Each question carries **4 marks**.

11. Plasmids are extrachromosomal entities in prokaryotic cells
    Origin of replication, cloning vectors, high copy number, antibiotic resistance genes.
    Egs: pBR 322, PUC 8 [U] [1]

12. Annotation and prediction of genes and genomes.
    Identifying proteins through the complex sequence similarity searches using protein sequence databases.
    Using bioinformatics tools and sequence motifs/regulatory elements genes can be clustered by function, and the co-expression characteristics can be determined.
    Identification of three-dimensional protein structures, molecular modeling, and folding to predict the possible function of proteins or other molecular structures, model behavior of molecules. [U] [1]

13. It consists of two major components, biotic or living components and nonbiotic or nonliving components. Biotic components include plants, animals, decomposers. Nonliving components include air, water, land. [U] [2]

14. Biodiversity hotspots are important for a balanced ecosystem. Biodiversity underpins all life on Earth. Without species, there would be no air to breathe, no food to eat, no water to drink. There would be no human society at all. [U] [3]

15. Genomics is an area within genetics that concerns the sequencing and analysis of an organism's genome. The genome is the entire DNA content that is present within one cell of an organism. It can be broadly classified into structural genomics and functional genomics. The former deals with the computational mapping of genomics elements like genes, regulatory sequences, repeats, polymorphsims etc, while the later focus on interpreting the functional roles, pathways, interactions etc of these genomic elements. [K] [4]

16. "Survival of the fittest" is a popular term that refers to the process of natural selection, a mechanism that drives evolutionary change Natural selection works by giving individuals who are better adapted to a given set of environmental conditions an advantage over those that are not as well adapted. In dangerous circumstances, only those individuals most adapted to their environment survive -and the world, with its limited food supply, fearsome predators, and devastating diseases is always a dangerous place. [U] [6]

17. Application of computational tools on molecular (biological) data including means to acquire, analyze, interpret, visualize and comprehend such data is bioinformatics.
The development and application of computational methods to analyse large collections of biological data, such as genetic sequences, cell populations or protein samples, to make new predictions using analytical methods, mathematical modelling and simulation to study biological, behavioral, and social systems. It encompasses frameworks, theories, algorithms, techniques and tools from mathematics, computer science and biology with the aim of understanding the significance of a variety of biological data. [An] [7]

18. In the central dogma of molecular biology, DNA, RNA, and proteins are related in the following ways:
DNA to RNA: DNA is transcribed into RNA by the enzyme RNA polymerase.
RNA to protein: RNA is translated into proteins.
Role of RNA: RNA carries genetic information from DNA to ribosomes, where proteins are synthesized.
There are three types of RNA that are necessary for protein synthesis:
Messenger RNA (mRNA): Contains the genetic information for a protein's amino acid sequence
Transfer RNA (tRNA): Matches amino acids to the corresponding mRNA sequences
Ribosomal RNA (rRNA): Makes up ribosomes, the organelles where proteins are synthesized
The central dogma of molecular biology describes the flow of genetic information in a biological system. It's often summarized as "DNA makes RNA makes protein".
[An][5]

(6 X 4 =24 Marks)

Part C

Essay Type Questions. Answer any **Two** questions.
Each question carries **8** marks**.**

19. Definition of Biotechnology- 1 mark
Application of Biotechnology in Agriculture with suitable examples – 2 marks
(GM crops, Genetic Engineering in agriculture, Golden Rice, Pest resistant Bt cotton etc)

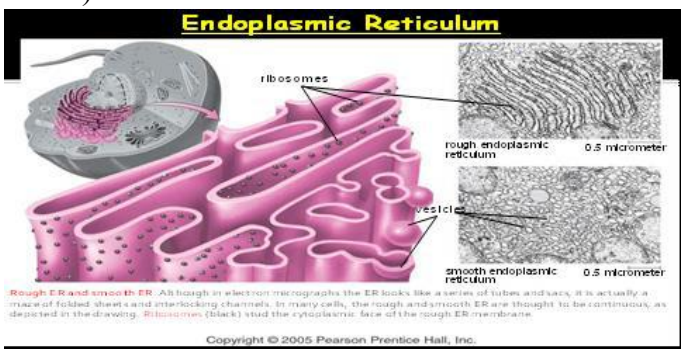
Application of Biotechnology in Medicine and Diagnosis with suitable examples- 2marks
(Vaccines,DNA Fingerprinting,Enzyme preparations, Gene Therapy,Diagnostic Kits)
Industrial applications- 1 mark
Environmental applications- 1 mark
Transgenic animals-1mark [K] [1]

20. Biological Databases. Definition- 1 mark
Primary, secondary and composite databases with two examples each- 3 marks
NCBI- database of databases.-1 mark
Other databases Literature databases, OMIM, KEgg and others.- 3 marks
(Name of database, the data they store and the significance of databases) [U] [2]

21. a)NMR (Nuclear Magnetic Resonance) **-** 1mark
What: Measures how atoms interact with magnetic fields.
How: Hydrogen atoms in molecules respond to magnetic fields, emitting signals.
Info: Molecular structure, chemical bonds, and dynamics. (2 marks)
Applications: (1 mark):- Drug Discovery: Identifies potential drug candidates, Material Science: Analyzes material properties, Food Safety: Detects contaminants.

b) X-RAY CRYSTALLOGRAPHY
What: Uses X-rays to visualize molecular structures.
How: X-rays scatter off electrons in crystallized molecules.
Info: 3D molecular structure, atomic arrangement. (1 mark each X 3)
Applications: (1 mark):- Protein Structure: Understands protein function, Drug Design: Develops targeted therapies. Vaccine Development: Identifies viral structures [A] [3]

22. **ENDOPLASMIC RETICULUM (ER) (3 marks)**
The endoplasmic reticulum (ER) is a network of membranes that plays a crucial role in protein synthesis and lipid metabolism. There are two types of ER: rough ER, which is studded with ribosomes and is involved in protein synthesis, and smooth ER, which lacks ribosomes and is involved in lipid metabolism and detoxification (4 marks)



4 marks for diagrammatic sketch [An] [5]

(2 X 8 = 16 Marks)

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